

API Usage Instructions & Payload Guide

Base URL

Live Railway Deployment:

<https://road-damage-detection-api-production.up.railway.app>

Endpoint 1: Vision-Only Detection (POST /detect)

Accepts an image file multipart upload, executes the fine-tuned RT-DETR model, and returns raw bounding boxes, non-COCO class labels (potholes, structural cracks, surface wear, manholes), and confidence scores.

cURL Request Command:

```
curl -X 'POST' \
  'https://road-damage-detection-api-production.up.railway.app/detect' \
  -H 'accept: application/json' \
  -H 'Content-Type: multipart/form-data' \
  -form 'file=@road_damage_sample.jpg'
```

Sample Response Payload:

```
{
  "status": "success",
  "total_detections": 2,
  "detections": [
    {
      "class_name": "pothole",
      "confidence": 0.8912,
      "bbox": [120.5, 340.2, 210.8, 415.6]
    },
    {
      "class_name": "crack",
      "confidence": 0.7643,
      "bbox": [45.0, 100.1, 180.4, 220.0]
    }
  ]
}
```

Endpoint 2: Natural-Language Reasoning (POST /reason)

Accepts an image file and a natural-language question. It routes the intent, runs vision inference, executes the confidence guardrail (≥ 0.35), and generates a structured plain-language response via Groq.

cURL Request Command:

```
curl -X 'POST' \
```

```
'https://road-damage-detection-api-production.up.railway.app/reason' \  
-H 'accept: application/json' \  
-H 'Content-Type: multipart/form-data' \  
-form 'file=@road_damage_sample.jpg' \  
-form 'question="How severe is the road damage shown in this picture?"'
```

Sample Response Payload (Success):

```
{  
  "status": "success",  
  "response": "Based on the visual analysis, the system detected 1 pothole with high confidence."  
}
```

Sample Response Payload (Guardrail Triggered / Low Confidence):

```
{  
  "status": "low_confidence",  
  "response": "The detection results are insufficient to answer confidently. Human verification is required."  
}
```